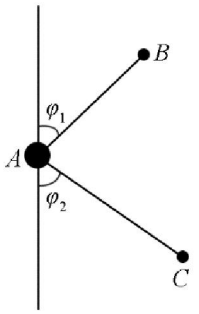
IV. For particles A, B, C, A is far heavier than B, C; B and C have equal masses. A attracts B, A attracts C, while B repels C. They move under these mutual forces. With suitable forces, the following motion is possible:

The motion has these properties:

A, B, C keep fixed relative positions and rotate about an axis in their plane, uniformly. Since A is far heavier than B and C, take the axis fixed through A.

The angles of AB, AC to the axis, φ_1, φ_2, differ; $0<\varphi_1<\frac{\pi}{2}$, $0<\varphi_2<\frac{\pi}{2}$.

The attractions are $f_{AB}=k|AB|^a$ for AB and, for AC, $f_a=k|AC|^a$; $|AB|$, $|AC|$ are the A–B, A–C distances. k is a proportionality constant. Ignore gravity. For which a can this motion occur?



V. (15 points) A huge iceberg breaks off an Antarctic ice shelf, drifting near a city. Use the iceberg as follows. (a) Seal ambient air in a variable-volume vessel. Cool the air at constant pressure to the iceberg temperature by contact. (b) Remove the vessel from the iceberg. At fixed volume, let its cold air absorb ambient heat to ambient temperature. (c) Eject air at fixed volume to run a generator. Repeat until all ice melts. The surroundings are at $T_a=293\text{K}$, the ice at $T_1=273\text{K}$. The available ice mass is $m=1.0\times10^{11}\,\text{kg}$. To estimate the electrical energy, the designer makes these assumptions and uses the following data:

1. Air is an ideal gas.